



Computer Society
VIT Student Branch



Smart Retail Supply Chain & Customer Intelligence Platform

1. Project Background

A mid-sized retailer in India operates 50+ stores, an e-commerce site, and a logistics network. Currently, their data is trapped in "silos" (separate systems like POS billing, web logs, and warehouse databases). This makes it impossible to see the full picture, leading to frequent stock-outs of popular items and a disconnected customer experience.

The Goal: Build an end-to-end "Data Hub" that gathers, cleans, and organizes this information for high-speed analysis and future AI use cases.

2. Technical Requirements

A. Data Ingestion (The "Pipelines")

You must build automated pathways to collect data from stores, warehouses, and the web.

- * Methods: Support both Batch (large chunks of daily data) and Near Real-Time (instant updates for online orders).

- * Resilience: The system must handle "Schema Evolution" (when data formats change slightly) and include automatic retries if a connection fails.

B. Transformation & Modeling (The "Organization")

Turn messy raw data into a structured format for the business.

- * Cleaning: Remove duplicates, fix missing values, and ensure data logic (e.g., no negative prices).

- * Tracking Changes (SCD): Ensure the system remembers history (e.g., if a customer moves from Mumbai to Delhi, we need a record of both).

- * The "Star/Snowflake" Schema: Organize data into central Fact tables (Sales, Inventory, Shipments) connected to Dimension tables (Products, Stores, Customers).

C. Storage, Access & Security (The "Foundation")

This is a critical engineering requirement. You must demonstrate how the data is stored and protected:

- * Optimized Storage: Use formats (like Parquet or Delta) that allow for fast analytical searching.

- * Partitioning Strategy: Organise data (e.g., by "Date" or "Region") so queries don't have to scan the entire database, saving time and cost.
- * Secure Access: Define how you would restrict data access so only authorized analysts can see sensitive information.

3. Analytics & Key Metrics

Once the data is organized, use it to calculate these specific Business KPIs:

- * Commercial: Daily/Monthly revenue, city-wise sales, and top-selling products.
- * Operations: Inventory Turnover Ratio, average delivery times, and seasonal demand trends.
- * Customer: New vs. Returning shoppers, Customer Lifetime Value (CLV), and "Market Basket" analysis (what items are bought together).

4. Expected Deliverables

To be successfully evaluated, candidates must submit:

1. Architecture Diagram: A visual map showing how data moves from the source to the final report.
2. Data Pipeline Code: Your Python, SQL, or Spark scripts used for moving and cleaning data.
3. The Analytical Dataset: A sample of your final, cleaned tables ready for a database.
4. Storage & Security Plan: A short write-up on your Partitioning strategy and Access controls.
5. Data Quality Documentation: Evidence of your automated checks (e.g., how you handled "impossible" values).
6. Analytics Scripts: The specific SQL or Python queries used to calculate the KPIs listed in Section 3.
7. Interactive Dashboard: A visual report (Power BI, Tableau, etc.) showing sales trends, inventory health, and delivery bottlenecks.

5. Sample Data Suggestions

You may use any public retail datasets to prove your solution works:

- * Kaggle: Search for "Retail Transactions" or "E-commerce behavior."
- * UCI Repository: "Online Retail" dataset.
- * Custom Data: You may use a script (like Faker) to generate realistic "mock" data, provided you explain the logic.